UNIT I – FUNDAMENTALS AND MATHEMATICAL FOUNDATIONS OF CRYPTOGRAPHY

Syllabus: Basic Security Goals, Types of Cryptographic Attacks, Security Services and Mechanisms, Cryptographic Methods, Integer Arithmetic, Modular Arithmetic, Matrices, Linear Congruence.

Textbook basis: Behrouz A. Forouzan, Introduction to Cryptography and Network Security. Chapter 1 covers security goals, attacks, services, mechanisms and cryptographic methods; Chapter 2 covers the mathematical foundations used in cryptography.

1. Introduction to Cryptography

Cryptography is a fundamental technology used to protect information in computer systems and networks. In the digital world, information may include passwords, financial transactions, academic records, business information, government information, medical records and authentication credentials.

The fundamental security objectives are Confidentiality, Integrity and Availability. These three objectives are commonly called the CIA Triad.

2. Basic Security Goals

2.1 Confidentiality

Confidentiality means protecting information from unauthorized access or disclosure. Only authorized users or entities should be able to access confidential information.

- A student's examination marks should be accessible only to authorized persons.
- A customer's bank account information should not be visible to unauthorized users.
- Passwords should not be disclosed to unauthorized persons.
- A confidential email should not be readable by an attacker.

Cryptography helps confidentiality mainly through encryption, which converts readable plaintext into an unintelligible ciphertext.

2.2 Integrity

Integrity means ensuring that information is not modified, deleted, inserted or altered by an unauthorized entity.

Example: If Alice sends 'Transfer Rs. 5,000' and an attacker changes it to 'Transfer Rs. 50,000', integrity has been compromised.

2.3 Availability

Availability means that information and services should be accessible to authorized users whenever required.

Example: An examination portal must remain accessible during registration and examination-related activities. A denial-of-service attack can threaten availability.

2.4 CIA Triad

Goal	Meaning	Main Question
Confidentiality	Prevent unauthorized disclosure	Who can see the information?
Integrity	Prevent unauthorized modification	Has the information been altered?
Availability	Ensure authorized access	Can authorized users access it when needed?

Memory aid: C – Keep it secret; I – Keep it correct; A – Keep it accessible.

3. Types of Security / Cryptographic Attacks

Forouzan's Chapter 1 introduces attacks according to the security goal they threaten and also classifies them as passive or active attacks.

3.1 Attacks Threatening Confidentiality

Snooping

Snooping is unauthorized access to or interception of data. An attacker attempts to observe or obtain information that is not intended for them.

Example: An attacker intercepts an unencrypted message and reads its contents.

Traffic Analysis

Traffic analysis occurs when an attacker monitors communication patterns even when message contents are protected. The attacker may study sender, receiver, frequency, timing and size of messages.

Traffic padding can make traffic analysis more difficult.

3.2 Attacks Threatening Integrity

Modification

Modification occurs when an attacker changes legitimate information after intercepting or accessing it.

Masquerading

Masquerading, or spoofing, occurs when an attacker pretends to be another legitimate entity.

Replaying

A replay attack occurs when an attacker captures a valid message and retransmits it later.

Repudiation

Repudiation occurs when one of the communicating parties later denies having performed an action or having sent/received a message.

3.3 Attack Threatening Availability – Denial of Service (DoS)

A Denial-of-Service attack attempts to make a system or service unavailable to legitimate users. An attacker may overload a server with requests or otherwise consume its resources.

3.4 Passive and Active Attacks

Feature	Passive Attack	Active Attack
Main objective	Obtain information	Modify or disrupt
Data changed?	Normally no	Yes, or system operation is

		affected
Main concern	Confidentiality	Integrity / Availability
Examples	Snooping, traffic analysis	Modification, masquerading, replay, repudiation, DoS
Detection	Often difficult	Usually comparatively easier

4. Security Services

A security service is a service designed to provide protection against security attacks and support security goals. The five security services described in the textbook are:

1. Data Confidentiality: Protects information against unauthorized disclosure.

2. Data Integrity: Protects information against unauthorized modification, insertion, deletion and replay.

3. Authentication: Establishes confidence about the identity of an entity or the origin of data.

4. Nonrepudiation: Prevents a sender or receiver from later denying a communication or transaction.

5. Access Control: Protects resources against unauthorized access.

5. Security Mechanisms

A security mechanism is a method used to implement a security service. The mechanisms described in the textbook include:

- Encipherment
- Data integrity mechanism
- Digital signature
- Authentication exchange
- Traffic padding
- Routing control
- Notarization
- Access control

Security Service	Examples of Supporting Mechanisms
Data confidentiality	Encipherment, routing control
Data integrity	Encipherment, digital signature, data integrity mechanism
Authentication	Encipherment, digital signature, authentication exchange
Nonrepudiation	Digital signature, data integrity, notarization
Access control	Access control mechanism

Important distinction: A security service describes WHAT protection is required; a security mechanism describes HOW that protection is implemented.

6. Cryptographic Methods

6.1 Symmetric-Key Encipherment

Symmetric cryptography uses the same secret key for encryption and decryption.

Plaintext → Encryption + Secret Key → Ciphertext

Ciphertext → Decryption + Same Secret Key → Plaintext

The communicating parties must possess or securely establish the shared secret key.

6.2 Asymmetric-Key Encipherment

Asymmetric cryptography uses two related keys: a public key and a private key. For confidential communication to Bob, Alice can encrypt using Bob's public key and Bob can decrypt using his private key.

Encryption: Plaintext + Bob's Public Key → Ciphertext

Decryption: Ciphertext + Bob's Private Key → Plaintext

6.3 Hashing

Hashing converts a variable-length message into a fixed-length message digest. The digest can be used as a check value for data integrity.

6.4 Cryptography and Steganography

Cryptography hides or transforms the meaning of a message. Steganography hides the existence of a message.

Cryptography	Steganography
Hides the meaning/content	Hides the existence of the message
Transforms plaintext into ciphertext	Embeds information inside a cover such as an image, audio or video

PART B – MATHEMATICAL FOUNDATIONS

Chapter 2 provides the mathematical foundation required for cryptography. The important topics are integer arithmetic, divisibility, GCD, Euclidean algorithm, Extended Euclidean algorithm, linear Diophantine equations, modular arithmetic, congruence, inverses, matrices, determinants, residue matrices and linear congruence.

7. Integer Arithmetic

7.1 Set of Integers – Z

$$Z = \{ ..., -3, -2, -1, 0, 1, 2, 3, ... \}$$

The set Z contains all positive integers, negative integers and zero.

7.2 Binary Operations

The common binary operations considered are addition, subtraction and multiplication. A binary operation takes two inputs and produces one output.

7.3 Integer Division

When an integer a is divided by a positive integer n, we obtain a quotient q and a remainder r.

$$a = q \times n + r$$

$$0 \leq r < n$$

Here, a is the dividend, n is the divisor, q is the quotient and r is the remainder.

Example: 157 = 12 × 13 + 1. Therefore q = 13 and r = 1.

7.4 Divisibility

If a = q × n for some integer q, then n divides a. This is written as n | a.

$$n \mid a \iff a = q \times n$$

Example: 4 | 32 because 32 = 4 × 8. But 8 does not divide 42 because 42 = 8 × 5 + 2.

8. Greatest Common Divisor (GCD)

The greatest common divisor of two positive integers is the largest integer that divides both. It is written as gcd(a, b).

Example: gcd(12, 140) = 4.

8.1 Euclidean Algorithm

The Euclidean algorithm is an efficient method for calculating GCD.

$$gcd(a, b) = gcd(b, r)$$

Example: Find gcd(36, 10).

$$36 = 10 \times 3 + 6$$

$$10 = 6 \times 1 + 4$$

$$6 = 4 \times 1 + 2$$

$$4 = 2 \times 2 + 0$$

$$\text{Therefore, } gcd(36, 10) = 2.$$

8.2 Extended Euclidean Algorithm

The Extended Euclidean Algorithm calculates the GCD and also finds integers s and t satisfying:

$$a \times s + b \times t = gcd(a, b)$$

This is particularly important for finding multiplicative inverses in modular arithmetic.

9. Linear Diophantine Equations

$$a \times x + b \times y = c$$

A linear Diophantine equation requires integer solutions for x and y.

Let d = gcd(a, b). If d does not divide c, there is no integer solution. If d divides c, integer solutions exist and there are infinitely many solutions.

Example: 21x + 14y = 35. Since gcd(21, 14) = 7 and 7 divides 35, solutions exist.

10. Modular Arithmetic

In modular arithmetic, we are mainly interested in the remainder obtained after division.

$$a \bmod n = r$$

$$0 \leq r < n$$

Here n is the modulus and r is the residue.

Examples:

$$27 \bmod 5 = 2$$

$$36 \bmod 12 = 0$$

$$-18 \bmod 14 = 10$$

$$-7 \bmod 10 = 3$$

10.1 Set of Residues – Z□

$$Z□ = \{0, 1, 2, \ldots, n - 1\}$$

Examples:

$$Z_2 = \{0, 1\}$$

$$Z_6 = \{0, 1, 2, 3, 4, 5\}$$

$$Z_{11} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

10.2 Congruence

$$a \equiv b \pmod{n}$$

This means that a and b have the same remainder when divided by n.

$$a \equiv b \pmod{n} \iff n \mid (a - b)$$

Examples:

$$2 \equiv 12 \pmod{10}$$

$$13 \equiv 23 \pmod{10}$$

$$-8 \equiv 12 \pmod{10}$$

10.3 Equality vs Congruence

Equality means two quantities are exactly the same. Congruence means they belong to the same residue class under a specified modulus.

$$5 = 5$$

$$5 \equiv 15 \pmod{10}$$

10.4 Residue Class

A residue class [a] modulo n contains all integers that are congruent to a modulo n.

$$[0] \bmod 5 = \{ \ldots, -10, -5, 0, 5, 10, \ldots \}$$

$$[1] \bmod 5 = \{ \ldots, -9, -4, 1, 6, 11, \ldots \}$$

11. Operations in Modular Arithmetic

$$(a + b) \bmod n = [(a \bmod n) + (b \bmod n)] \bmod n$$

$$(a - b) \bmod n = [(a \bmod n) - (b \bmod n)] \bmod n$$

$$(a \times b) \bmod n = [(a \bmod n) \times (b \bmod n)] \bmod n$$

Examples:

$$(14 + 7) \bmod 15 = 6$$

$$(7 - 11) \bmod 13 = 9$$

$$(7 \times 11) \bmod 20 = 17$$

12. Additive Inverse

In Z□, a and b are additive inverses if their sum is congruent to zero modulo n.

$$a + b \equiv 0 \pmod{n}$$

Example: In Z_{10}, 4 and 6 are additive inverses because 4 + 6 = 10 ≡ 0 (mod 10).

$$\text{Additive inverse of } a \text{ in } Z_n = n - a \quad (\text{for } a \neq 0)$$

13. Multiplicative Inverse

Two numbers a and b are multiplicative inverses modulo n if their product is congruent to 1 modulo n.

$$a \times b \equiv 1 \pmod{n}$$

Example: In Z_{10}, 3 and 7 are multiplicative inverses because 3 × 7 = 21 ≡ 1 (mod 10).

13.1 Condition for Existence of a Multiplicative Inverse

$$a^{-1} \text{ exists modulo } n \iff \gcd(a, n) = 1$$

Therefore, a and n must be relatively prime.

Example: 8 has no multiplicative inverse modulo 10 because gcd(8, 10) = 2, not 1.

13.2 Finding a Multiplicative Inverse Using the Extended Euclidean Algorithm

Suppose we want b^{-1} modulo n. If the Extended Euclidean Algorithm gives:

$$s \times n + t \times b = 1$$

then:

$$t \times b \equiv 1 \pmod{n}$$

$$\text{Therefore, } b^{-1} \equiv t \pmod{n}$$

Example: The inverse of 11 modulo 26 is 19.

$$11 \times 19 = 209 \equiv 1 \pmod{26}$$

14. Z□ and Z□*

$$Z□ = \{0, 1, 2, \ldots, n - 1\}$$

Z□ contains all least residues. Every element has an additive inverse, but not every element has a multiplicative inverse.

Z□* contains the elements of Z□ that have multiplicative inverses.

$$Z_{10} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

$$Z_{10}^* = \{1, 3, 7, 9\}$$

15. Matrices

A matrix is a rectangular arrangement of elements. If it has l rows and m columns, its size is l × m. The element a_{ij} is located in row i and column j.

$$A = [\ a_{ij}\ ]$$

Example:

$$A = [\ 2\ \ 4\ \ 6\ ;\ \ 1\ \ 3\ \ 5\ ]$$

The example is a 2 × 3 matrix.

15.1 Types of Matrices

- Row matrix – contains one row.
- Column matrix – contains one column.
- Square matrix – has the same number of rows and columns.
- Identity matrix – square matrix with 1s on the main diagonal and 0s elsewhere.
- Zero matrix – all elements are zero.

15.2 Matrix Equality

Two matrices are equal if they have the same dimensions and all corresponding elements are equal.

15.3 Matrix Addition and Subtraction

Matrices can be added or subtracted only when they have the same dimensions.

$$A + B = [\ a_{ij} + b_{ij}\ ]$$

$$A - B = [\ a_{ij} - b_{ij}\ ]$$

15.4 Matrix Multiplication

If A is an l × m matrix and B is an m × p matrix, then AB is an l × p matrix.

$$A(l \times m) \times B(m \times p) = C(l \times p)$$

Important rule: The number of columns of the first matrix must equal the number of rows of the second matrix.

15.5 Scalar Multiplication

$$2 \times [\ 1\ \ 3\ ;\ 4\ \ 5\ ] = [\ 2\ \ 6\ ;\ 8\ \ 10\ ]$$

16. Determinant

The determinant is defined only for square matrices.

For a 2 × 2 matrix:

$$A = [\ a\ \ b\ ;\ c\ \ d\ ]$$

$$\det(A) = a \times d - b \times c$$

Example:

$$A = [\ 3\ \ 4\ ;\ 2\ \ 5\ ]$$

$$\det(A) = (3 \times 5) - (4 \times 2) = 7$$

17. Matrix Inverse

The multiplicative inverse of a square matrix A is a matrix A^{-1} such that:

$$A \times A^{-1} = A^{-1} \times A = I$$

For ordinary real-valued matrices, the inverse exists when det(A) ≠ 0.

18. Residue Matrices

A residue matrix is a matrix whose elements belong to Z_n. Matrix operations are performed using modular arithmetic.

A residue matrix has a multiplicative inverse if its determinant has a multiplicative inverse in Z_n.

$$\text{Residue matrix A is invertible modulo n} \iff \gcd(\det(A), n) = 1$$

19. Matrix Congruence

Two matrices A and B are congruent modulo n if they have the same dimensions and corresponding elements are congruent modulo n.

$$A \equiv B \pmod{n}$$

$$a_{ij} \equiv b_{ij} \pmod{n} \quad \text{for every } i \text{ and } j$$

20. Linear Congruence

$$a \times x \equiv b \pmod{n}$$

Let d = gcd(a, n).

- If d does not divide b, there is no solution.
- If d divides b, there are exactly d solutions modulo n.

General solution when d divides b:

$$x = x_0 + k \times (n / d), \quad \text{where } k = 0, 1, \ldots, d - 1$$

20.1 Example – No Solution

$$10x \equiv 2 \pmod{15}$$

$$\gcd(10, 15) = 5$$

Since 5 does not divide 2, there is no solution.

20.2 Example – Multiple Solutions

$$14x \equiv 12 \pmod{18}$$

$$\gcd(14, 18) = 2$$

Since 2 divides 12, there are exactly two solutions.

$$14x \equiv 12 \pmod{18}$$

$$7x \equiv 6 \pmod{9}$$

$$7^{-1} \equiv 4 \pmod{9}$$

$$x \equiv 6 \times 4 \equiv 6 \pmod{9}$$

$$\text{Solutions modulo } 18\text{: } x = 6, 15$$

20.3 Linear Congruence with a Constant

$$3x + 4 \equiv 6 \pmod{13}$$

$$3x \equiv 2 \pmod{13}$$

$$3^{-1} \equiv 9 \pmod{13}$$

$$x \equiv 2 \times 9 \equiv 18 \equiv 5 \pmod{13}$$

Therefore x ≡ 5 (mod 13).

21. System of Linear Congruences

A set of linear congruences with the same modulus can be represented using matrices.

$$A\ X \equiv B \pmod{n}$$

If A has a multiplicative inverse modulo n, then:

$$X \equiv A^{-1}\ B \pmod{n}$$

Here A is the coefficient matrix, X is the variable column matrix and B is the right-hand-side column matrix.

22. Important Textbook Example – System of Congruences

$$3x + 5y + 7z \equiv 3 \pmod{16}$$

$$x + 4y + 13z \equiv 5 \pmod{16}$$

$$2x + 7y + 3z \equiv 4 \pmod{16}$$

$$[\ 3\ \ 5\ \ 7\ ;\ 1\ \ 4\ \ 13\ ;\ 2\ \ 7\ \ 3\ ]\ [\ x\ ;\ y\ ;\ z\ ] \equiv [\ 3\ ;\ 5\ ;\ 4\ ] \pmod{16}$$

$$x \equiv 15 \pmod{16},\quad y \equiv 4 \pmod{16},\quad z \equiv 14 \pmod{16}$$

23. Connections Among the Mathematical Topics

The mathematical topics form a logical sequence:

1. Integer arithmetic
2. Divisibility
3. Greatest Common Divisor (GCD)
4. Euclidean Algorithm
5. Extended Euclidean Algorithm
6. Modular arithmetic
7. Congruence and residue classes
8. Additive and multiplicative inverses
9. Matrices and determinants
10. Residue matrices
11. Linear congruence
12. Systems of linear congruences

The Extended Euclidean Algorithm is particularly important because it is used to obtain modular multiplicative inverses, which in turn are required for solving many linear congruences and matrix operations modulo n.

24. Cryptography Connection with Mathematics

Mathematical Concept	Importance in Cryptography
Integer arithmetic	Basic numerical operations
Divisibility	Number-theoretic operations
GCD	Determines relative primality
Euclidean Algorithm	Efficient GCD computation
Extended Euclidean Algorithm	Finding modular inverses

Modular arithmetic	Core arithmetic in many cryptographic algorithms
Congruence	Represents equivalence modulo n
Z□	Set of residues used in cryptographic calculations
Z□*	Elements having multiplicative inverses
Matrices	Represent transformations and systems
Determinant	Helps determine matrix invertibility
Residue matrices	Matrix operations modulo n
Linear congruence	Used to solve cryptographic equations

25. Quick Revision Table

Topic	**Key Idea / Formula**
Confidentiality	Prevent unauthorized disclosure
Integrity	Prevent unauthorized modification
Availability	Ensure authorized access
Snooping	Unauthorized access/interception
Traffic analysis	Monitoring communication patterns
Modification	Unauthorized alteration
Masquerading	Impersonation
Replay	Retransmission of captured message
Repudiation	Denial of communication/action
DoS	Makes service unavailable
Passive attack	Mainly observes
Active attack	Changes/disrupts
Security services	Confidentiality, integrity, authentication, nonrepudiation, access control
Security mechanisms	Encipherment, integrity, digital signature, authentication exchange, traffic padding, routing control, notarization, access control
Symmetric cryptography	Same secret key
Asymmetric cryptography	Public/private key pair
Hashing	Fixed-length digest
Integer division	$a = q \times n + r$
Divisibility	$n \mid a$ if $a = q \times n$
GCD	Greatest common divisor
Euclidean algorithm	Efficient GCD
Extended Euclidean	$a \times s + b \times t = \gcd(a,b)$
Diophantine equation	$a \times x + b \times y = c$
Modulo	$a \bmod n = r$
Residue set	$Z_n = \{0, ..., n-1\}$
Congruence	$a \equiv b \pmod n$
Additive inverse	$a + b \equiv 0 \pmod n$
Multiplicative inverse	$a \times b \equiv 1 \pmod n$
Inverse condition	$\gcd(a,n) = 1$
Matrix multiplication	Columns of first = rows of second
Determinant	Defined for square matrices
Matrix inverse	$A \times A^{-1} = I$
Residue matrix inverse	$\gcd(\det(A),n) = 1$
Linear congruence	$a \times x \equiv b \pmod n$
Solution condition	$\gcd(a,n)$ divides b
System of congruences	$A X \equiv B \pmod n$

26. Important Questions for Examination

1. Define confidentiality, integrity and availability.
2. Explain the CIA Triad with suitable examples.
3. Explain snooping and traffic analysis.
4. Explain modification, masquerading, replay and repudiation.
5. What is a Denial-of-Service attack?
6. Differentiate passive and active attacks.
7. List and explain the five security services.
8. List and explain the security mechanisms.
9. Differentiate security services and security mechanisms.
10. Explain symmetric-key and asymmetric-key cryptography.
11. Differentiate cryptography and steganography.
12. What is hashing?
13. Define divisibility and explain its notation.
14. Calculate GCD using the Euclidean algorithm.
15. Explain the Extended Euclidean Algorithm.
16. Solve a linear Diophantine equation.
17. Define modular arithmetic and explain the modulo operator.
18. Define Z□ and residue class.
19. Differentiate equality and congruence.
20. Explain additive and multiplicative inverses.
21. State and explain the condition for a multiplicative inverse.
22. Find a modular inverse using the Extended Euclidean Algorithm.
23. Explain matrix multiplication and its dimension rule.
24. Find the determinant of a 2 × 2 matrix.
25. Explain residue matrices and the condition for their inverse.
26. Solve ax ≡ b (mod n).
27. Explain how a system of linear congruences can be represented using matrices.

28. Practice Problems

- Find the quotient and remainder when 527 is divided by 23.
- Determine whether 17 divides 289.
- Find gcd(88, 220), gcd(300, 42), gcd(24, 320), and gcd(401, 700).
- Use the Extended Euclidean Algorithm to find gcd(291, 42) and the corresponding coefficients.
- Find 22 mod 7, 140 mod 10, −78 mod 13, and 0 mod 15.
- Calculate (273 + 147) mod 10.
- Calculate (125 × 45) mod 10.
- Find the additive inverse of every element in Z_{20}.
- Find the multiplicative inverse pairs in Z_{20}.
- Find the inverse of 7 modulo 18.
- Find the inverse of 11 modulo 26.
- Solve 3x ≡ 4 (mod 5).
- Solve 4x ≡ 4 (mod 6).
- Solve 256x ≡ 442 (mod 60).
- Solve 3x + 5 ≡ 4 (mod 5).
- Find the determinant of a given 2 × 2 matrix.
- Determine whether a residue matrix modulo 10 is invertible.

- Solve a given system of two or three linear congruences using the matrix method.

29. Final Conceptual Summary

The first part of Unit I establishes why cryptography is required, while the second part establishes the mathematics needed to understand cryptographic algorithms.

Security Requirements → Security Goals → Attacks → Security Services → Security Mechanisms → Cryptographic Methods

Integer Arithmetic → Divisibility → GCD → Euclidean Algorithm → Modular Arithmetic → Inverses → Matrices → Linear Congruence

A strong understanding of GCD, the Euclidean Algorithm, modular arithmetic, modular inverses, matrices and linear congruences will make the subsequent study of cryptographic algorithms considerably easier.

Reference

Behrouz A. Forouzan, Introduction to Cryptography and Network Security, relevant material from Chapter 1 and Chapter 2 (Mathematics of Cryptography).